

Interactive Digital Art and Design SE608

Proof of Concept - Concept Alignment Year IV

Student Name:	Jessica Curran	Student Number:	C00300855
Working Title:	Pixel Paradise		
PoC Date:	02/10/2026	PoC Version:	Version 1
How to use this document:			
1. Project Theme / Problem:		Pixel Paradise is a 3D game which will be focused on creating low poly environments and implementing them into a game engine to create an exploration game. The player will travel through a series of different environments, each environment will contain an animal that the player must find and collect. The aim will be to work your way through the world, exploring all environments and trying to collect all of the animals. The project will explore how a visually appealing 3D environment can be combined with simple interactive gameplay mechanics to create an engaging exploration game. A key part of the project will be to demonstrate the process of creating the environments and assets in Crocotile 3D and implementing them into Godot to create a visually appealing and engaging game.	
2. Project Aim:		To create an engaging 3D exploration game that combines detailed environments with simple gameplay mechanics to create an enjoyable and immersive experience for the player.	
3. Project Objectives:		- Design and create a wide range of 3D environments and assets in Crocotile 3D and Aseprite - Develop a playable exploration game in Godot that allows the player to move through and explore the environments - Implement an animal collection mechanic where the player must locate and collect an animal in each environment	
4. Target Users / Audience:		The primary target audience is players who enjoy casual or relaxing exploration games. The game is intended to provide a relaxing exploration experience where the player can discover new environments and locate animals within them. The game may also appeal to people who enjoy collecting games where the main objective is to just explore the world and collect items, rather than games that are complex or competitive.	
5. Main User Requirements / Use Cases:		- Start and enter the game world - Control a player character and explore the environments - Navigate through the different areas of the game world - Identify and locate the animals in different environments - Interact with and collect the animals - Receive indication that the animal has been collected - Explore all environments and collect all animals to complete the game	

6. Required Features / Functional Requirements:	Essential features • Playable 3d player character • Player movement and camera control • Basic sounds/background music • Multiple completed 3D environments • 3D environments created in Crocotile 3D • Importing assets into Godot • Collision and navigations within the environments • Animal characters within the environments • Animal interaction and collection mechanics • Indication when animal has been collected • Playable final game Optional/improvement features • More advanced sound effects and background music • More complex animal animations • Rigging pixel art player character • Animal collection counter and information • Main menu and settings																												
7. Proposed Tools and Technologies:	<table><tr><th>Tool / Technology</th><th>Purpose in Project</th><th>Feature / Objective Supported</th><th>Why Suitable?</th></tr><tr><td>Aseprite</td><td>Designing and creating tilesets to import into Crocotile</td><td>Tileset, player and animal character creation</td><td>Aseprite is a 2D pixel art creation software which will work perfectly alongside Crocotile which is a 3D pixel art software</td></tr><tr><td>Crocotile 3D</td><td>Designing and creating 3D environments, buildings, props and other assets</td><td>Environment and asset creation</td><td>I want to create 3D pixel art assets, and this is perfect for that, it allows the environments to be created efficiently and compatible with Godot</td></tr><tr><td>Godot Engine</td><td>Building and coding the playable game and implementing the 3D assets</td><td>Player movement, interaction and gameplay</td><td>Provides the tools required to create a playable 3D game implanting assets from Crocotile 3D</td></tr><tr><td>GitHub</td><td>Project backup</td><td>Project management</td><td>Allow me to track changes made to the project and provide a backup for my project.</td></tr><tr><td>Spine 2D</td><td>Rigging characters</td><td>Character rigging/ animations</td><td>Spine 2D will work well for rigging pixel art characters created in Aseprite</td></tr></table>					Tool / Technology	Purpose in Project	Feature / Objective Supported	Why Suitable?	Aseprite	Designing and creating tilesets to import into Crocotile	Tileset, player and animal character creation	Aseprite is a 2D pixel art creation software which will work perfectly alongside Crocotile which is a 3D pixel art software	Crocotile 3D	Designing and creating 3D environments, buildings, props and other assets	Environment and asset creation	I want to create 3D pixel art assets, and this is perfect for that, it allows the environments to be created efficiently and compatible with Godot	Godot Engine	Building and coding the playable game and implementing the 3D assets	Player movement, interaction and gameplay	Provides the tools required to create a playable 3D game implanting assets from Crocotile 3D	GitHub	Project backup	Project management	Allow me to track changes made to the project and provide a backup for my project.	Spine 2D	Rigging characters	Character rigging/ animations	Spine 2D will work well for rigging pixel art characters created in Aseprite
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	A playable exploration game	Player movement, collision and item collection	Godot/C#	Player movement throughout the environments	Player can navigate and explore the game world																				
	Animal collection	Animal interaction and collection system	Godot/C#	Player approaching the animal and collecting it	Player can successfully locate and collect the animals																				
	Implement 3D assets	Import 3D assets into game	Crocotile 3D/Godot	Imported and working 3D assets inside Godot game	3D assets function and work correctly within game																				
9. PoC Demonstration / Evidence:		For my PoC demonstration, I will create a small test environment in Crocotile 3D to show that I can properly use this software to create 3D assets. I will then implement this test environment into Godot. Once I have the test environment added, I will create a very simple player movement and collision test and add a collectable to show that I will be able to complete all of the basic requirements of this project to create a functional game.																							
10. Technology Suitability / Alternatives:		<table><tr><th>Requirement</th><th>Proposed Approach / Tool</th><th>Alternative Considered</th><th>Reason for Final Choice</th></tr><tr><td>3D environment creation</td><td>Crocotile 3D</td><td>Maya/Blender</td><td>Crocotile 3D is best suited as I like the pixel art style and how the software works</td></tr><tr><td>Game engine</td><td>Godot</td><td>Unity</td><td>I chose Godot because it is very compatible with Crocotile 3D and very easy to implement the 3D assets</td></tr><tr><td>Gameplay coding</td><td>C#</td><td>GDScript</td><td>C# is compatible with Godot and is suitable for simple gameplay mechanics</td></tr><tr><td></td><td></td><td></td><td></td></tr></table>				Requirement	Proposed Approach / Tool	Alternative Considered	Reason for Final Choice	3D environment creation	Crocotile 3D	Maya/Blender	Crocotile 3D is best suited as I like the pixel art style and how the software works	Game engine	Godot	Unity	I chose Godot because it is very compatible with Crocotile 3D and very easy to implement the 3D assets	Gameplay coding	C#	GDScript	C# is compatible with Godot and is suitable for simple gameplay mechanics				
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11. Scope - Core vs Optional Features:		<table><tr><th>Feature</th><th>Priority</th><th>Reason / Objective Supported</th><th>Include in PoC?</th></tr><tr><td>A range of playable environments</td><td>Must have</td><td>3D environment creation and player exploration</td><td>Yes</td></tr><tr><td>Player movement</td><td>Must have</td><td>Allows the player to explore the environments</td><td>Yes</td></tr><tr><td>Animals to discover and collect</td><td>Must have</td><td>Provides the main gameplay interaction</td><td>Yes</td></tr></table>				Feature	Priority	Reason / Objective Supported	Include in PoC?	A range of playable environments	Must have	3D environment creation and player exploration	Yes	Player movement	Must have	Allows the player to explore the environments	Yes	Animals to discover and collect	Must have	Provides the main gameplay interaction	Yes				
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	Crocotile 3D assets	Must have	Shows the main artistic focus of the project	Yes																		
	Gameplay audio	Must have	Improves the atmosphere of the game	Yes																		
12. Feasibility Assessment:	<i>Assess whether the project is realistically achievable. Consider: Technical - can the required features be built? Time - can a meaningful version be completed? Skills - can the required skills be developed in time? Resources - are software, hardware, data and participants available? Evaluation - can success be demonstrated or measured?</i> <table><tr><th>Area</th><th>Question to Consider</th><th>Assessment / Evidence</th></tr><tr><td>Technical</td><td>Can the required features be built?</td><td>Yes. The required features are quite limited and consist mainly of 3D assets, player movement and simple interaction.</td></tr><tr><td>Time</td><td>Can a meaningful version be completed?</td><td>Yes. The project will focus on core environments, player movement and collection mechanics first before additional features are added.</td></tr><tr><td>Skills</td><td>Can the required skills be developed in time?</td><td>Yes. The project is using my strengths of 3D art and asset creation. Basic programming knowledge will be required but it will be limited to my level and past experience.</td></tr><tr><td>Resources</td><td>Are software, hardware and data available?</td><td>Crocotile 3D, Aseprite and Godot are the main software I will be using that I already have installed.</td></tr><tr><td>Evaluation</td><td>Can success be demonstrated or measured?</td><td>Success can be demonstrated through a playable exploration game with basic character movement, collision and interaction</td></tr></table>				Area	Question to Consider	Assessment / Evidence	Technical	Can the required features be built?	Yes. The required features are quite limited and consist mainly of 3D assets, player movement and simple interaction.	Time	Can a meaningful version be completed?	Yes. The project will focus on core environments, player movement and collection mechanics first before additional features are added.	Skills	Can the required skills be developed in time?	Yes. The project is using my strengths of 3D art and asset creation. Basic programming knowledge will be required but it will be limited to my level and past experience.	Resources	Are software, hardware and data available?	Crocotile 3D, Aseprite and Godot are the main software I will be using that I already have installed.	Evaluation	Can success be demonstrated or measured?	Success can be demonstrated through a playable exploration game with basic character movement, collision and interaction
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14. PoC Outcome / Decision:	Select one outcome and justify the decision below: ☐ Proceed - the concept and proposed approach are suitable. ☐ Proceed with changes - the concept is feasible, but changes are required. ☐ Reconsider - the current concept or approach requires significant revision. Justification: [Explain the decision in 3-5 sentences.]												
15. Required Changes Before Full Development:	List the specific changes that must be made as a result of the PoC. Examples: change technology, reduce scope, replace an input method, redefine a feature, modify the user flow, add a Plan B, or strengthen the technical component.												
16. Next Steps:	List the next 3-5 concrete development tasks in priority order. Each task should follow logically from the PoC decision and the revised project direction.												
17. Supporting Material / References:	Provide links or references to supporting research, technical documentation, comparable systems, design references, GitHub repository, screenshots, diagrams or short demonstration videos used to justify the PoC decisions.												
Student declaration: The information in this Proof of Concept reflects the current proposed direction of the FYP and the evidence available at this stage. Signed: Jessica Curran Date: 02/10/2026													
For Office / Supervisor Use Only													
PoC reviewed:	[Supervisor notes]												
Main issues / conditions:	[Supervisor notes]												
Approved direction / required changes:	[Supervisor notes]												
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